

SET A

1. The Midpoint Line Algorithm (MPL) offers the following advantages over the Digital Differential Analyzer (DDA) algorithm:
 - a. MPL uses only integer addition, subtraction, and bit shifting operations, making it faster and more efficient.
 - b. DDA requires floating-point operations which may suffer from rounding errors.

2.

- a. Here, the given endpoints are– P1 (11, 20), P2(2, 12)

$$dx = x_2 - x_1 = 2 - 11 = -9$$

$$dy = y_2 - y_1 = 12 - 20 = -8$$

$$\text{Now, } |dx| = 9, |dy| = 8$$

$$dx < 0, dy < 0 \text{ and } |dx| > |dy|$$

Thus, the line lies in **Zone = 4**

- b. Zone = 6 represents, $dx > 0, dy < 0$ and $|dy| > |dx|$

(X, Y) will be converted to (-Y, X) [Negate then Swap]

$$(5, 9) \rightarrow (-9, 5)$$

[Note: The order of zone conversion from Z to 0 and 0 to Z are reversed.

zone 6 to zone 0 conversion: Negate, then Swap

zone 0 to zone 6 conversion: Swap, then Negate]

3. P1 = (2, 3) , P2 = (19, 7)

$$dx = 19 - 2 = 17, dy = 7 - 3 = 4 \text{ [Thus, it is in Zone 0]}$$

$$d(\text{init}) = 2dy - dx = 2 * 4 - 17 = -9$$

$$\Delta s(\text{NE}) = 2dy - 2dx = 2 * 4 - 2 * 17 = -26$$

$$\Delta s(\text{E}) = 2dy = 2 * 4 = 8$$

Pixel	X	Y	d	NE / E	Δs	updated d
(2, 3)	2	3	-9	E	8	$-9+8=-1$
(3, 3)	3	3	-1	E	8	$-1+8=7$
(4, 3)	4	3	7	NE	-26	$7-26=-19$
(5, 4)	5	4	-19	E	8	$-19+8=-11$
(6, 4)	6	4	-11	E	8	$-11+8=-3$
(7, 4)	7	4	-3	E	8	$-3+8=5$
(8, 4)	8	4				

SET B

1. The Midpoint Line Algorithm (MPL) offers the following advantages over the Digital Differential Analyzer (DDA) algorithm:
 - a. MPL uses only integer addition, subtraction, and bit shifting operations, making it faster and more efficient.
 - b. DDA requires floating-point operations which may suffer from rounding errors.

2.

- a. Here, the given endpoints are— P1 (9, 19), P2(2, 13)

$$dx = x_2 - x_1 = 2 - 9 = -7$$

$$dy = y_2 - y_1 = 13 - 19 = -6$$

$$\text{Now, } |dx| = 7, |dy| = 6$$

$$dx < 0, dy < 0 \text{ and } |dx| > |dy|$$

Thus, the line lies in **Zone = 4**

- b. Zone = 2 represents, $dx < 0, dy > 0$ and $|dy| > |dx|$

(X, Y) will be converted to (Y, -X) [Negate then Swap]

$$(5, 9) \rightarrow (9, -5)$$

[Note: The order of zone conversion from Z to 0 and 0 to Z are reversed.

zone 2 to zone 0 conversion: Negate, then Swap

zone 0 to zone 2 conversion: Swap, then Negate]

3. P1 = (1, 4) , P2 = (17, 11)

$$dx = 17 - 1 = 16, dy = 11 - 4 = 7 \quad [\text{Thus, it is in Zone 0}]$$

$$d(\text{init}) = 2dy - dx = 2 * 7 - 16 = -2$$

$$\Delta s(\text{NE}) = 2dy - 2dx = 2 * 7 - 2 * 16 = -18$$

$$\Delta s(\text{E}) = 2dy = 2 * 7 = 14$$

Pixel	X	Y	d	NE / E	Δs	updated d
(1, 4)	1	4	-2	E	14	-2+14=12
(2, 4)	2	4	12	NE	-18	12-18=-6
(3, 5)	3	5	-6	E	14	-6+14=8
(4, 5)	4	5	8	NE	-18	8-18=-10
(5, 6)	5	6	-10	E	14	-10+14=4
(6, 6)	6	6	4	NE	-18	-18+4=-14
(7, 7)	7	7				